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标题	Sustainable Mizoroki – Heck Cross-Coupling Using a Pd(II)-Polymer as Precatalyst in 1 - Butanol
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摘要（原文节选）

This work describes the application of a palladium-containing polymeric precatalyst (Pd/PECIm) for the sustainable Mizoroki – Heck (MH) cross-coupling reaction, focusing on the use of green solvents. Building on previous research in which Pd/PECIm was successfully applied to the Suzuki – Miyaura reaction, the cross-coupling between iodobenzene and ethyl acrylate was initially performed in water and 2-propanol (iPrOH). While low yields were obtained in water or iPrOH alone, good yields were obtained in a mixture of both ⋯

数据表 · 1（12 行，含数值行 12 行）

Optimization of Reaction Conditions for MH Reaction between iodobenzene and Ethyl Acrylate Catalyzed by Pd/PECIm

entry	Pd/mol %	T/ ° C	t/h	solvent	yield/%
1	0.5	80	24	iPrOH	17
2	0.5	80	24	iPrOH/H2O	66
3	0.5	80	24	H2O	67
4	1.0	80	24	iPrOH/H2O	69
5	1.5	80	24	iPrOH/H2O	70
6	2.0	80	24	iPrOH/H2O	69
7	0.5	90	24	iPrOH/H2O	88
8	1.0	90	24	iPrOH/H2O	87
9	0.5	80	24	EtOH	79
10	0.5	95	1.5	1-BuOH	95
11	-	95	1.5	1-BuOH	0
12	-	95	1.5	1-BuOH	0

脚注（逐字）：aReaction conditions: iodobenzene (0.5 mmol), ethyl acrylate (1.0 mmol), Et3N (1.0 mmol), solvent (1.0 mL), and Pd/PECIm (ranging from 2.5 to 10 mg).bYield determined by GC/MS with a calibration curve for ethyl cinnamate.cA total of 2.5 mg of PECIm was used.dNo additional components were added.

正文对应句（逐字）：The MH reaction between iodobenzene and ethyl acrylate was selected to evaluate the catalytic performance of Pd/PECIm under varying conditions of solvent, reaction time, and catalyst loading (Table). Guided by the solvent selection principles outlined by Prat et al., isopropanol (iPrOH), classified

数据表 · 2（17 行，含数值行 17 行）

Scope of MH Coupling Reaction Catalyzed by Pd/PECIm

entry	X	R 1	R 2	t/h	yield/%	TON	TOF/h − 1
1	I	H	CO2CH3	2	91	176	88
2	I	H	CO2CH2CH3	1.5	95	182	121
3	I	H	CO2(CH3)3	2	95	182	91
4	I	H	C6H5	24	91	176	7
5	I	4-CH3O	C6H5	24	73	141	6
6	I	4-CH3CO	C6H5	48	66	127	3
7	I	4-CH3O	CO2CH2CH3	3	90	173	58
8	I	4-NO2	CO2CH2CH3	4	90	173	43
9	I	2-CH3	CO2CH2CH3	2	95	182	91
10	I	4-CH3O	CO2C(CH3)3	2	92	178	89
11	I	2-CH3	CO2C(CH3)3	2	91	176	88
12	Br	H	CO2C(CH3)3	48	15	29	0.6
13	Br	4-NO2	CO2CH2CH3	4	90	173	43
14	Br	4-CH3CO	C6H5	48	56	108	2
15	Cl	H	CO2C(CH3)3	56	0	0	0
16	I	H	4-NC5H5	24	46	89	4
17	I	H	CO2CH2CH3	2	93	172	86

脚注（逐字）：aReaction conditions: aryl halide (0.5 mmol), alkene (1.0 mmol), Et3N (1.0 mmol), n-BuOH (1.0 mL), Pd/PECIm (2.5 mg), 95 ° C.bIsolated yield.cTOF was calculated based on the total amount of Pd atoms in Pd/PECIm.dPerformed at 120 ° C.ePerformed on a 50× scale.

正文对应句（逐字）：The MH reaction between iodobenzene and ethyl acrylate was selected to evaluate the catalytic performance of Pd/PECIm under varying conditions of solvent, reaction time, and catalyst loading (Table). Guided by the solvent selection principles outlined by Prat et al., isopropanol (iPrOH), classified

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